Analytical Derivation of Free-Surface Velocity in a Tank with Deviated Walls

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1 Problem Setup

We consider a tank with deviated walls, characterized by a bottom diameter d_1, a top diameter d_2, and a total height h. The fluid drains through an orifice of diameter d_0 located at the bottom. The instantaneous water-column height is denoted by z, measured from the bottom (see Fig. 1).

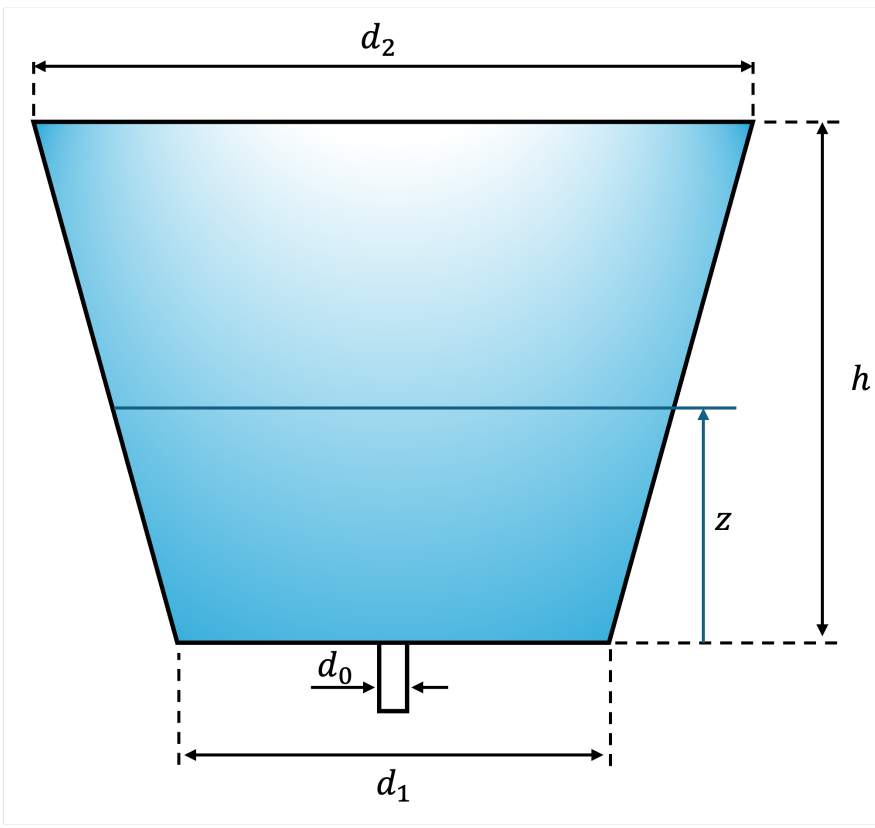

Figure 1: Tank geometry and notation: top diameter d_2, bottom diameter d_1, height h, orifice diameter d_0, and instantaneous water-column height z.

Let

$$A_1(z) = \frac{\pi}{4}\, d^2(z)$$

be the tank cross-sectional area at height z, and

$$A_2 = \frac{\pi}{4}\, d_0^2$$

be the orifice area.

We define:

- $V_1(z)$: velocity of the free surface at height z,
- $V_2(z)$: exit velocity at the orifice.

1.1 assumptions

- The fluid is incompressible (constant density).
- The viscosity of the fluid is negligible (inviscid flow).

2 Governing Equations

2.1 Bernoulli Equation

Applying Bernoulli between the free surface (1) and the orifice (2):

$$\frac{1}{2}\rho V_1^2 + \rho g z = \frac{1}{2}\rho V_2^2, \tag{1}$$

since both points are at atmospheric pressure.

2.2 Continuity

Conservation of mass gives

$$A_1(z) V_1 = A_2 V_2. \tag{2}$$

From (2) we can express the exit velocity in terms of the free-surface velocity:

$$V_2 = \frac{A_1(z)}{A_2}\, V_1. \tag{3}$$

3 Free-Surface Velocity

Substituting (3) into Bernoulli (1):

$$\frac{1}{2}\rho V_1^2 + \rho g z = \frac{1}{2}\rho \left(\frac{A_1(z)}{A_2} V_1\right)^2, \tag{4}$$

$$\Rightarrow \quad g z = \frac{1}{2} V_1^2 \left[\left(\frac{A_1(z)}{A_2}\right)^2 - 1\right]. \tag{5}$$

Solving for the free-surface velocity $V_1(z)$ yields

$$\boxed{V_1(z) = \sqrt{\frac{2gz}{\left(\frac{A_1(z)}{A_2}\right)^2 - 1}}} \tag{6}$$

This equation provides the instantaneous free-surface velocity, as there are two parts which depend on z it integration is a little hard, so I use python to get the answer.